Air Quality EDA Report By Vamshi Aitharaju

Firstly, for my Exploratory Data Analysis, When I looked at all of the categorical and Quantitative data, I Have only one Quantitative and rest Categorical has More of Meta Data such as Measure and measure info, Geo Location type and Geo Location Name. So, I wanted to create at least one meaningful variable for my EDA, so after reading my data for a bit I felt like I can extract Community District from Geo Location Name from all the 3 sheets (Annual average Air Quality, Winter Air Quality and Summer Air Quality), so that I can use it for Plotting a chart for more detailed analysis. Then After I did Descriptive Stats for all of my 3 sheets based on Pollutants because each pollutant is different and there is no point in mixing up them together. So I Here my observations, There were no data about O3 in Winter and Annual but only in Summer which highest average of 30 PPB while NO2 is 16PPB, But winter has very high NO2 average with 25PPM and Annual Average shows NO2 is higher in than PM2.5, So in my research I found that the Lowest recorded PM2.5 is highly dangerous than O3 and No2. The Dangerous PM2.5 is stable in every season with slight change.

Approach and Methods Used to perform EDA

My Approach and Methods used to Perform EDA is

1. Composition
2. Comparison
3. Relations
4. Outliers
5. Pivots

1.Compostion (Share of Pollutants): In this part I wanted to show which pollutants are sharing their composition to form whole data. I have used 100% stacked bar as **First Chart** to show how much are PM2.5, No2 and O3 are contributing in annual, winter and Summer. Where we can see NO2 share is higher than PM2.5. But in Summer PM2.5 is higher which is highly dangerous and better to take some precautions. An **Insight** was been taken from Professor Kristin, As O3 data is available only in summers and not in winter and annual, Professor recommended to remove it, which is completely agreeable as O3 data is not constant, their can be data inaccuracy.

2.Comparison (Difference across places and seasons): In comparison I have compared the Pollutant Values over the years and also across the places. So, the **Second Chart** was Bar Chart of Annual Avg Pollutant Value by Place but her I have utilized the first step where I extract the CD, so I have added Slicer to CD so now I can clearly show which Community Dist. has more pollutant. So, IN CD 5 Mid Town has Highest Pollution whereas CD3 Tottenville and Great Kill has less Pollution of 9.39. The **Third Chart** is Line Chart of Summer Air Quality Trends By Year 2009–2023, where it is clear that Ozone has higher than PM2.5 and NO2 and increasing year by year but Both NO2 and PM2.5 are going slowly down almost Equally. **The Forth Chart** is Bar Chart of Average Pollutant Concentrations by Winter Season (2009–2023), where it seen that No2 is always higher than PM2.5, some years NO2 is double than PM2.5, But slowly Both Pollutant were coming down and were almost same in last 3 years. The Fifth Chart is Clustered Column Chart of Annual Avg Pollutant by Year, where is same NO2 is higher than Pm2.5 ever year and both came slowly down and increased a bit in last year. **Note**: I have removed grand totals of every pollutant, as is not that useful for my analysis.

3. Relations (Correlation between pollutants): Sixth Chart is Scatter Plot: PM2.5 vs NO_2 (Annual Average), where I can see dots spread everywhere and no pattern seen and R Square is 0.0061, which say Both pollutants are moving independently and not corelated to each other. Coming to **Seventh Chart** For the scatter plot of PM2.5 vs O_3 (Summer), The dots are scattered in middle and R Square is

0.0001, so just like Sixth Chart, there is no correlation between PM2.5 and O3 and Both are moving Independently.

4.Outliers (distribution and extremes): So, for the Distribution I have selected the Histograms, So the **Eighth Chart** is Histogram For NO2 (Annual Average), Most of the Values are in between 15 and 27 with Bin of 18-21 being Highest Bar. The histogram is Right Skewed which means that most of readings and values are in Mid-range, but there no that many extremes pollute events. The **Ninth Chart** is Histogram for PM2.5 (Annual Average), Most of the values are in-between 5-11 with 6.3 – 7 Bin being tallest bar, Her the histogram is same as previous one with Right Skewed, which means both histograms are positively skewed. For the Extremes I have selected Box plots, so the **Tenth Chart** is Boxplot for NO_2 (Annual vs summer vs Winter) Here I have overserved that in winter NO_2 is very high compared to summer and annual, median is 25 and some values even go above 40 which are outliers, so winter is more unstable. Summer NO_2 is very low and close together, so variation is less. Annual is in middle but closer to summer So winter is the reason why the average goes high. So, for the **Eleventh Chart** is Boxplot for PM2.5 (Annual vs summer vs Winter). This show PM2.5 is same in all seasons, but in summer the median is high and spread also wider. Annual is in between. Outliers are there in all but not that extreme like NO_2. So PM2.5 is more stable compare to NO_2, even though it is still very dangerous pollutant. An **Insight** has been Given by Professor Kristin here, The Suggestion was to compare the Same Pollutants by seasons rather than comparing different pollution Just like I did in tenth and eleventh chart.
Hypothesis Testing: As I have Already done hypothesis testing for No2 and Pm2.5 of Annual average. So, the p-value came out as 0 which is less than 0.05, which means the result is highly significant. It is clear PM2.5 and NO_2 averages are not equal. NO_2 is much higher than PM2.5, and the p-value proves this difference is real. So we need do predations for both pollutants separately. Then after I did hypothesis testing for winter and summer along with box plots and I could see anything, so next I have proceeded with the Professor Kirstin's insight, I have proceeded hypothesis testing for winter summer No2 and winter summer PM2.5 and the findings are very interesting. **Hypothesis Testing for Winter vs Summer PM2.5** So the t-test for PM2.5 between winter and summer shows a P-value of 0.91, which is greater 0.05. That means there is no difference in PM2.5 values between winter and summer. The averages are almost same which means both seasons behave very similar and P-value it is not significant, so we can combine winter and summer PM2.5 together for prediction. **Hypothesis Testing for Winter vs Summer NO2:** The t-test for NO_2 between winter and summer p-value close 0. That means there is a significant difference. Winter means and summer mean is different, so we need to treat winter and summer NO_2 separately and do prediction individually as p-value is less than 0.05.

How I dealt outliers or anomalies in the data

1. I found few outliers in my data but mostly found high outliers in winter NO2 data, few in PM2.5 but I have removed them because they are real pollutant events, so I have focused more on averages so the extreme points don't change the main outcome results.
2. I tried to create a Bubble chart of PM2.5, NO2 and O3 but, it did not came out that good, all the bubbles were of same size, which is not adding any valuable insight for my analysis so I have not included the bubble chart

Here by I (Vamshi Aitharaju) declare that I have done this report writing on my own and is ready for Review. Thank You.

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